

Getting Started

Set up your own accounts and connect the toolkit — a one-time process, about 10 minutes.

This toolkit finds, reviews, and downloads satellite imagery over an area you draw, and helps you plan field visits around future satellite passes. It uses three outside services, each needing its own free account: **Google Earth Engine** (imagery search), **NASA Earthdata** (real archive downloads), and optionally **GitHub** (to back up your project). None of this costs money for normal personal/research use.

■■ **Watch out** — **EE_PROJECT** is intentionally blank in the notebooks, and the app's project field starts empty. Earth Engine projects are tied to one Google account, so there is no shared default that could work for everyone — you must create your own (Step 2) and fill it in before anything connects to Earth Engine. Leaving it blank raises a clear error telling you exactly that.

Step 1 — Install Python packages

From the project folder, install everything the toolkit needs:

```
pip install -r requirements.txt
```

Step 2 — Create your own Earth Engine project

- 1 Go to `code.earthengine.google.com/register` and sign in with any Google account.
- 2 Choose **"Unpaid usage"** (free, for research/personal/education) unless you specifically need a commercial license.
- 3 When asked for a Cloud project, either create a new one or pick an existing Google Cloud project — write down its **Project ID** (a short lowercase-hyphenated string like `my-name-12345`, shown under the project's display name — not the display name itself).
- 4 Registration automatically enables the Earth Engine API on that project — no separate step needed.

■ **Tip** — Free Google accounts (Gmail) work fine. You do not need a company or paid Google Cloud billing account for normal-sized research use.

Step 3 — Point the toolkit at your project

Fill your Project ID into the blank in each place it's needed:

- In every notebook (`1_AOI_Selection.ipynb` through `4_Future_Overpass_Forecast.ipynb`): find the line `EE_PROJECT = ""` near the top and put your Project ID between the quotes.
- In the dashboard app: type your Project ID into the **"Earth Engine project"** box in the sidebar before clicking Connect.

- For downloads, also set `EARTHDATA_USERNAME = ""` in `3_Download_Filtered_Scenes.ipynb` (Step 5 below), and `GITHUB_USERNAME = ""` in `6_Push_To_GitHub.ipynb` if you use the sync step.

Step 4 — Sign in (no code to copy/paste)

Run the setup cell in any notebook, or click "**Connect to Earth Engine**" in the app's sidebar. The first time, your browser opens to a Google sign-in page:

- 1 Sign in with the same Google account you registered in Step 2.
- 2 Click **Allow** on the permissions screen.
- 3 The browser tab shows a success message — switch back to your notebook/app, it finishes automatically. Nothing to copy or type.

■ **Tip** — This only happens once per computer. After the first successful sign-in, a credential file is cached and every future run just connects instantly.

Step 5 — NASA Earthdata Login (only needed for Download)

Some products (Sentinel-1, NISAR, Sentinel-6, and the HLS harmonized products) come from NASA's real data archive rather than Earth Engine, and downloading them needs a free Earthdata account:

- 1 Register at `urs.earthdata.nasa.gov` (free, a few minutes).
- 2 You'll enter this username/password directly in the Download step when you're ready to download — it's never saved to a file.

See the **Download Guide** for a note on one extra one-time step ASF DAAC requires (authorizing an application) before Sentinel-1/NISAR downloads work.

Step 6 — (Optional) GitHub, for backing up your project

Only needed if you want to push your project to your own private GitHub repository — see the **GitHub Sync Guide** for the full walkthrough (you'll need a free GitHub account and a Personal Access Token, not your GitHub password).

Step 7 — Run it

Two ways to use the toolkit, both use the exact same underlying code:

- **All-in-one dashboard** (recommended for first-time use): run `streamlit run app.py` and open the link it prints (usually `http://localhost:8501`).
- **Notebooks**: open `1_AOI_Selection.ipynb` and run the cells top to bottom, then `2_Sentinel_Search.ipynb`, and so on in order.

■ **Tip** — Stuck on a specific step? Each stage has its own short guide — open the Help tab in the dashboard, or the matching PDF in `docs/pdf/`.